CALCULUS · UNIT 4: APPLICATIONS OF DERIVATIVES

Package Design Studio

Use optimization, related rates, and curve analysis to design packaging for a juice company.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Fresh Squeeze Juice is launching a new drink and needs packaging that uses as little material as possible. The bottling plant fills cone-shaped tanks, and the marketing team wants to understand the sales curve.

You will use derivatives to design the package, control the filling process, and analyze sales.



Can Volume

355 cm³

(12 fl oz)

Box Sheet

30 cm by 20 cm

cardboard

Learning Goals

- I can solve optimization problems and justify extrema.
- I can solve related-rates problems.
- I can analyze a function with first and second derivatives.

Project Timeline

When	What you will do
Day 1	Optimize the Can
Day 2	Cone Tank
Day 3	Sales Analysis
Final	Design Proposal + reflection

Part 1: Optimize the Can

Day 1 · A can holds 355 cm^3. Minimize the surface area $A = 2\pi r^2 + 2\pi rh$.

Design Data Cards

Open Box	Can	Tank
Cut squares of side x	$V = 355\ cm^3$	Inverted cone
from a 30 × 20 sheet	minimize surface area	height 6 m, radius 2 m

Fill Rate	Sales	Cost
2 m^3 per minute	$S(t) = -t^3 + 12t^2 + 60t$	Aluminum 0.02¢ per cm^2
	thousand cans, t months	

1. Write h in terms of r and A as a function of r alone.

2. Find $A'(r)$ and the critical point.

3. Find h and compare h with the diameter. Justify that this is a minimum.

4. Real cans have $r \approx 3.3$ cm and $h \approx 10.4$ cm. How much more aluminum do they use (in cm^2)?

Part 2: Cone Tank

Day 2 · Inverted cone: height 6 m, radius 2 m. Fill rate 2 m^3/min.

5. Use similar triangles to write r in terms of h and V in terms of h.

6. Find $\frac{dh}{dt}$ when h = 3 m.

7. Find how fast the radius of the juice surface is increasing at h = 3.

8. Find how fast the surface area of the juice top is increasing at h = 3.

Part 3: Sales Analysis

Day 3 · $S(t) = -t^3 + 12t^2 + 60t$, $0 \le t \le 12$.

9. Find intervals of increase/decrease and concavity, and the inflection point.

10. Interpret the inflection point for the marketing team.

11. Use the MVT on [0, 10] to show that sales grew at exactly 80 thousand cans/month at some time. Find it.

Final Product: Design Proposal

Your final product must include:

- Optimal can with justification and cost comparison
- Cone-tank related rates
- Full sales curve analysis
- MVT application
- Design proposal

Reflection

R1. Why do real cans not use the optimal shape?

R2. What is the difference between where S is maximum and where S′ is maximum?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Optimization, related rates, and curve analysis are set up, solved, and justified correctly. Goes beyond the minimum with a creative, realistic choice.	Optimization, related rates, and curve analysis are set up, solved, and justified correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes diagrams, derivative work, justifications, and recommendations. Polished, organized, and visually engaging.	Proposal includes diagrams, derivative work, justifications, and recommendations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.